

Tri-mode Capable Audio Power Amplifier

Approach, Design & Functionality

Group 19

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Introduction

A tri-mode capable audio power amplifier is a power amplifier that can drive two speakers in three distinct configurations.

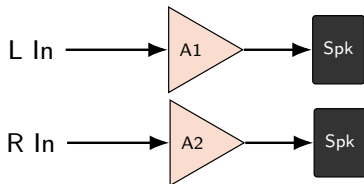
- **Mono Mode** – The Left and Right channels from the stereo input are summed and amplified to drive both speakers with the same audio signal.
- **Stereo Mode** – The Left and Right channels from the stereo input are individually amplified. Each channel drives only one speaker.
- **Bridge Mode** – Achieves a theoretical four-fold power output by driving only one speaker utilizing both power amplifiers.

Device Specifications

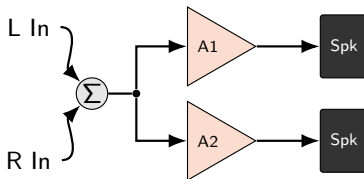
- **Supply Voltage** - $\pm 15V$
- **Maximum Input Voltage** - $1V_{peak}$
- **Maximum Output Voltage** - $3.5V_{peak}$ (From one channel)
- **Maximum Output Current** - $1A_{peak}$ (From one channel)
- **Frequency Response** - 20Hz - 20kHz
- **Speaker Load** - $4\Omega, 5W$

Tri-Mode Routing Block Diagrams

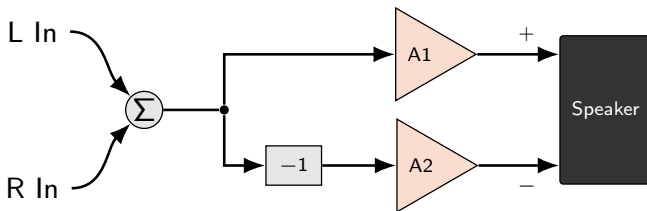
1. Stereo Mode



2. Mono Mode

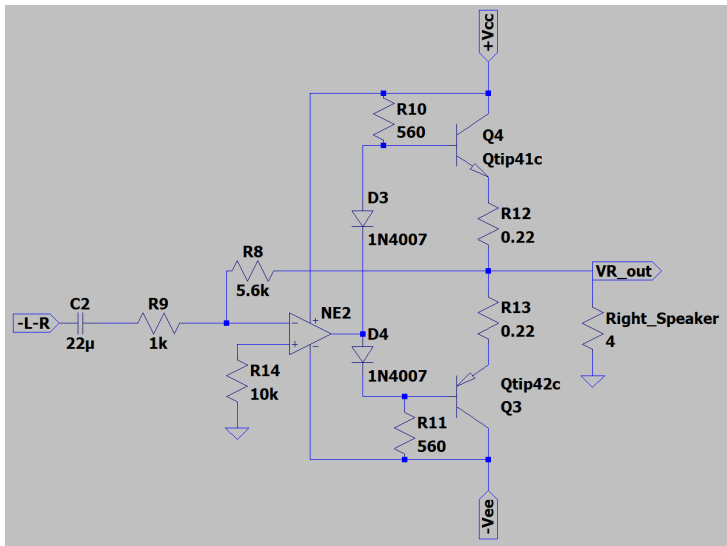


3. Bridge Mode



System Architecture & Approach

Class-AB power Amplifier



System Architecture & Approach (Continued)

Class-AB power Amplifier

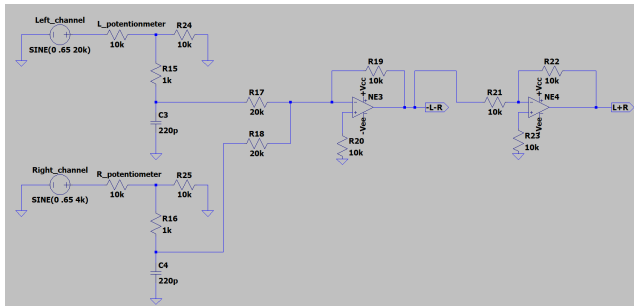
- **Voltage Stage:** The OpAmp provides the required voltage gain at the output. Adjusted by varying the feedback resistor.
- **Current Stage:** The complementary pair of power BJTs provides the current need to drive the speaker directly from the power line.

Circuit Operation

- OpAmp feedback system compares the input and output voltages with the set gain.
- OpAmp output current is fed into the bases of the 2 BJTs.
- 2 pn junction diodes make sure both BJTs are in active region at all times, overcoming crossover distortion.
- 0.22Ω emitter resistor protects the circuit from thermal runaway.
- 560Ω across collector and base ensures diodes are always forward biased.

System Architecture & Approach (Continued)

Left and Right Channel Summer & Invertor



Circuit Operation

- The two channel voltages are summed and inverted with a gain of **0.5**, to ensure the output won't be clipped.
- Above output is inverted again to obtain $L+R$ signal.

Operational Amplifier

Requirements

- Should be a **low noise** OpAmp.
- Must have a sufficient slew rate for 20kHz operation.

$$SR \geq |V_{pp} \cdot j\omega e^{j\omega t}| = 7 \times 2\pi \times 20 \times 10^3 = 0.879 \text{ V} / \mu\text{s}$$

- A rail-to-rail output across the operational bandwidth is preferred.
- Must be operational under $\pm 15\text{V}$ dual power supply.

Selection: NE5532

- Low noise OpAmp designed for audio operations.
- 9 V / μs Slew Rate.
- Supply voltage range: $\pm 3\text{V} - \pm 20\text{V}$

Component Selection (Continued)

Power Transistors

Requirements

- Must be a **complimentary pair** of NPN and PNP BJTs: To ensure similar performances in both output half cycles.
- Must support a continuous current of at least 1.5A. And a maximum power rating of 3W.
- Must have a heat sink to dissipate the heat generated.

Selection: TIP41 NPN & TIP42 PNP

- Have similar characteristics and performance.
- Can supply a continuous 6A current and has a power rating of 65W.
- TO-220 package provides a built in heat sink.

Biasing Diodes

Requirements

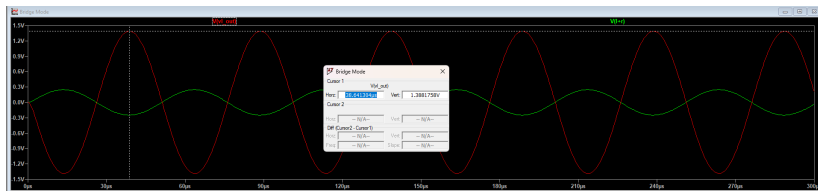
- Must have a forward voltage drop of 2V: To bring the power transistors to the on state ($V_{BE(on)} = 2V$).
- Must support a continuous current of at least 1A.

Selection: 1N4007 (connected 2 in series)

- Each has a forward voltage drop of 1.1V (In series: 2.2V).
- Have an average forward current of 1A.
- Readily available.

Simulation & Functionality (LTSpice)

Gain Verification



- Verified the OpAmp feedback circuit provide the voltage gain set by the resistor pairs. Above has a ≈ 5.6 gain achieved using $5.6k\Omega$ and $1k\Omega$ resistor pair.

Elimination of Crossover Distortion

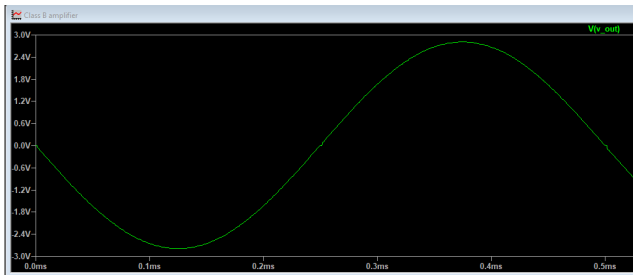
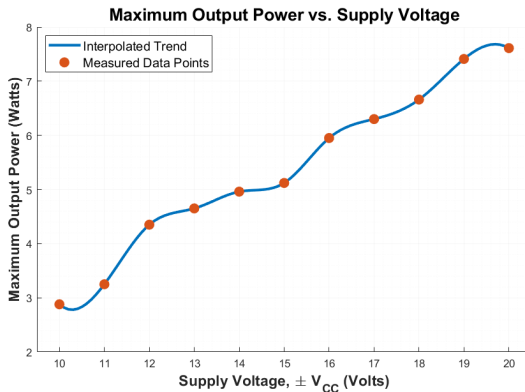


Figure: Result when biasing diodes are not present

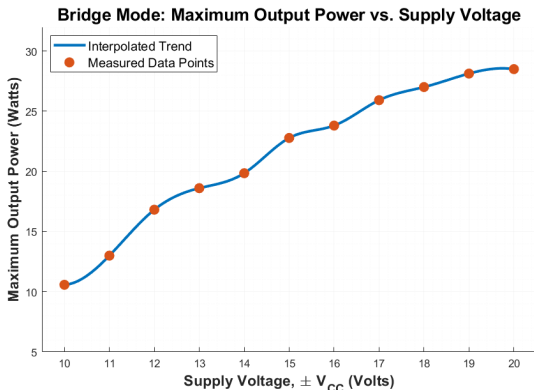
- Verified that the OpAmp complimented with the biasing diodes eliminate the crossover distortion occur in a pure class-B amplifier.

All the following tests are performed on a 4Ω load



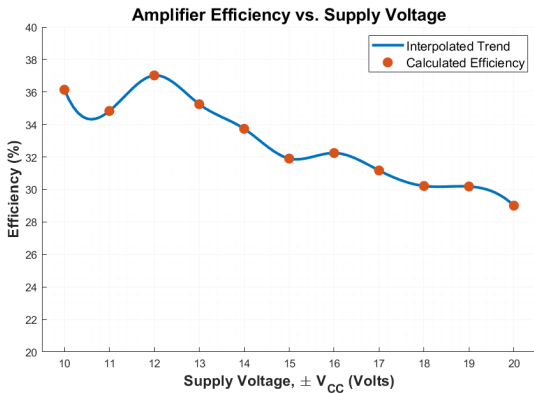
- The Supply voltage range was determined to be around 15V (typical) while satisfying power requirement for NE5532.

Testing & Analysis Cont...



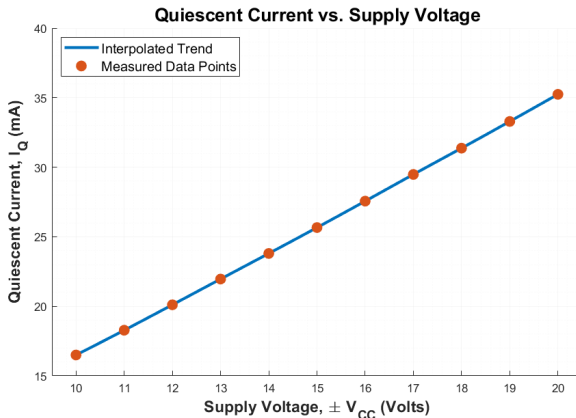
- Verified that the output power quadruples in the Bridge Mode.

Testing & Analysis Cont...



- The maximum efficiency was around 37%.

Testing & Analysis Cont...



- Current through the biasing diodes when idle.

Testing & Analysis Cont...

- Measured Amplifier Bandwidth: 23.8 kHz
- Measured Amplifier Slew Rate: $1.25 \text{ V}/\mu\text{s}$ $0.879 \text{ V}/\mu\text{s}$
- No harmonic distortions were present.

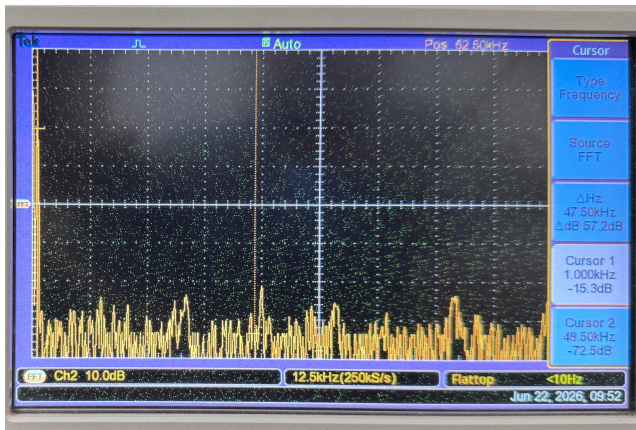


Figure: Output signal FFT of 1kHz input signal

References I

- [1] Douglas Self. *Audio Power Amplifier Design Handbook*, 4th Edition.
- [2] ALPHA Lab. “How does it work — Amplifier mode — what is the bridge mode in power amplifier.” YouTube, Apr. 14, 2022. [Online]. Available:
https://www.youtube.com/watch?v=Yaw14gN_WiQ
- [3] ElectroBOOM. “Making an Audio Power Amplifier (Drive a Sub Woofer).” YouTube, May 31, 2013. [Online]. Available:
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